B. LAKSHMI ANJALI

192311344

1. Software Defect Identification in an ATM System

Aim: To examine the specifications of an ATM system and identify software defects through testing.

Bug ID	Bug Description	Values Entered / Action Done	Expected Result	Actual Result	Status: Pass / Fail
TC01	Invalid PIN	Enter an incorrect PIN	Display an error message	Error displayed	Fail
TC02	Insufficient balance	Withdraw ₹10,000 when balance is ₹5,000	Reject transaction	Transaction rejected	Fail
TC03	Invalid withdrawal amount	Enter ₹-500	Reject invalid amount	Invalid amount rejected	Fail
TC04	Exceed daily limit	Withdraw more than the daily limit	Display limit exceeded message	Limit message displayed	Fail
TC05	Cash not dispensed	Complete transaction but cash is not dispensed	Reverse transaction or refund amount	Amount refunded	Fail
TC06	Repeated incorrect PIN	Enter incorrect PIN three times	Block card temporarily	Card blocked	Fail
TC07	Empty PIN field	Submit without entering PIN	Request PIN	Validation message displayed	Fail
TC08	Network failure	Disconnect network during transaction	Handle failure and maintain correct balance	Error displayed	Fail

Result: Software defects and potential failure scenarios in the ATM system were identified and documented.

2. Positive Test Scenarios and Test Cases for VIANA Portal

Aim: To analyze the requirements of the VIANA portal and prepare positive test cases to verify valid operations.

Bug ID	Bug Description	Values Entered / Action Done	Expected Result	Actual Result	Status: Pass / Fail
TC01	Valid login	Enter valid username and password	Login successful	Dashboard displayed	Pass
TC02	Student registration	Enter all valid details	Student registered successfully	Registration successful	Pass
TC03	Search student record	Enter a valid student ID	Display student details	Student details displayed	Pass
TC04	Update student details	Modify phone number with valid data	Details updated successfully	Details updated	Pass
TC05	View academic records	Select a valid semester	Display academic records	Records displayed	Pass
TC06	Download report	Click download for an available report	Report downloaded	Report downloaded	Pass
TC07	Logout	Click the logout button	Session terminated	Login page displayed	Pass

Result: Positive test cases were prepared to verify that the VIANA portal performs its intended functions when valid inputs are provided.

3. Negative Test Scenarios and Test Cases for VIANA Portal

Aim: To identify how the VIANA portal handles invalid inputs and unauthorized actions.

Bug ID	Bug Description	Values Entered / Action Done	Expected Result	Actual Result	Status: Pass / Fail
TC01	Invalid login	Enter incorrect username or password	Display an error message	Error displayed	Pass
TC02	Empty login fields	Submit without username and password	Display required-field messages	Validation displayed	Pass
TC03	Invalid student ID	Search using a nonexistent student ID	Display record not found	Record not found displayed	Pass
TC04	Invalid email	Enter `student123.com`	Reject invalid email format	Validation displayed	Pass
TC05	Missing registration details	Submit an incomplete registration form	Request required information	Error displayed	Pass

TC06	Unauthorized access	Student attempts to access an administrator page	Deny access	Access denied	Pass
TC07	Invalid phone number	Enter letters in the phone field	Reject invalid phone number	Validation displayed	Pass
TC08	Invalid semester	Select an unsupported semester	Display an appropriate error	Error displayed	Pass

Result: Negative test cases were prepared to verify that the VIANA portal correctly rejects invalid data and unauthorized requests.

4. Positive Test Scenarios and Test Cases for E-Commerce Applications

Aim: To analyze the requirements of an e-commerce application and verify its functionality using valid inputs.

Bug ID	Bug Description	Values Entered / Action Done	Expected Result	Actual Result	Status: Pass / Fail
TC01	User login	Enter valid username and password	Login successful	Login successful	Pass
TC02	Product search	Search for `Laptop`	Display matching products	Products displayed	Pass
TC03	Add product to cart	Select an available product and click Add to Cart	Product added to cart	Product added	Pass
TC04	Update quantity	Change product quantity to 2	Cart total updated correctly	Total updated	Pass
TC05	Apply valid coupon	Enter a valid discount coupon	Discount applied	Discount applied	Pass
TC06	Place order	Enter valid delivery details and confirm order	Order placed successfully	Order confirmation displayed	Pass
TC07	Successful payment	Enter valid payment details	Payment processed successfully	Payment successful	Pass
TC08	Track order	Enter a valid order ID	Display order status	Status displayed	Pass

Result: Positive test cases were prepared to verify that the e-commerce application works correctly with valid inputs.

5. Negative Test Scenarios and Test Cases for E-Commerce Applications

Aim: To analyze the requirements of an e-commerce application and verify how it handles invalid inputs, failed payments, and unexpected user actions.

Bug ID	Bug Description	Values Entered / Action Done	Expected Result	Actual Result	Status: Pass / Fail
TC01	Invalid login	Enter an incorrect password	Display an error message	Error displayed	Pass
TC02	Empty registration fields	Submit without required details	Display validation messages	Validation displayed	Pass
TC03	Invalid email format	Enter `user123.com`	Reject invalid email	Error displayed	Pass
TC04	Negative product quantity	Enter quantity as `-2`	Reject negative quantity	Quantity rejected	Pass
TC05	Out-of-stock product	Attempt to purchase an unavailable item	Display out-of-stock message	Message displayed	Pass
TC06	Invalid coupon	Enter `INVALID50`	Reject coupon code	Coupon rejected	Pass
TC07	Failed payment	Enter invalid payment details	Reject payment and show an error	Payment rejected	Pass
TC08	Empty cart checkout	Click Checkout without products	Prevent checkout	Checkout prevented	Pass
TC09	Incomplete address	Submit an incomplete delivery address	Request missing details	Validation displayed	Pass
TC10	Invalid quantity exceeding stock	Order more items than are available	Reject the unavailable quantity	Quantity rejected	Pass

Result: Negative test cases were prepared to verify that the e-commerce application rejects invalid inputs and handles failed operations appropriately.